

Sinian Zhang

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EDUCATION

University of Minnesota, Twin Cities

Ph.D. Student in Biostatistics

Minneapolis, US

Sep 2024 – Jun 2029 (Expected)

Dissertation Advisors: [Sandra Safo](#) and [Jue Hou](#) (Division of Biostatistics and Health Data Science)

Also Supervised by: [Ju Sun](#) (Computer Science & Engineering)

Renmin University of China

B.S. in Statistics

Beijing, China

Sep 2020 – Jun 2024

RESEARCH INTERESTS

Statistically principled methods for **transferable and trustworthy machine learning and deep learning**, with applications spanning biomedical, imaging, and engineering domains.

- **Transfer Learning** – estimators that keep their statistical guarantees when distributions shift or evidence must be pooled across sites, spanning distributional transfer, federated causal estimation, and reinforcement learning.
- **Uncertainty Quantification** – deciding when a model should abstain rather than guess, via selective prediction with explicit risk–coverage trade-offs.
- **Biomedical & Health Data Science** – computational and AI methods for electronic health records, biomedical knowledge graphs, medical imaging, and multi-platform biomarker data.

PUBLICATIONS

*Equal contribution.

Peer-Reviewed Publications

1. Sinian Zhang*, Chongwei Chen*, Guanchen Li, Ju Sun. Ensemble Selective Classification. *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. [📄 Code](#)
2. Gaoxiang Luo, Yifan Wu, Sinian Zhang, Aryan Deshwal, Ju Sun. Aligning Language Models with Selective Prediction. *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. Also at *ICML 2026 Workshop on Agents in the Wild: Safety, Security, and Beyond*. [📄 Paper](#) [📄 Code](#)
3. Jingxin Wang*, Sinian Zhang*, Yun Liang, Yibo Lin, Pengpeng Ren, Runsheng Wang, Weikang Qian. GUALO: A Generalizable Uncertainty-Aware AI Agent for Logic Optimization. *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2026.
4. Elzbieta Jodlowska-Siewert, Yunhui Chen, Sinian Zhang, Jia Li, Robert Dellavalle, Rui Zhang, Jue Hou. Antidiabetic Drug Associations With Heart Failure Outcomes: Real-World Evidence Study Using Electronic Health Records. *JMIR Diabetes* 11 (2026): e85083. [📄 Paper](#)
5. Kaicheng Zhang*, Sinian Zhang*, Doudou Zhou, Yidong Zhou. Wasserstein Transfer Learning. *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. [📄 Paper](#) [📄 Code](#)
6. Yongkang Xiao, Sinian Zhang, Yi Dai, Huixue Zhou, Jue Hou, Jie Ding, Rui Zhang. DrKGC: Dynamic Subgraph Retrieval-Augmented LLMs for Knowledge Graph Completion across General and Biomedical Domains. *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 16432–16445. [📄 Paper](#)

7. Feiqing Huang, Jue Hou, Ningxuan Zhou, Kimberly Greco, Chenyu Lin, Sara Morini Sweet, Jun Wen, Lechen Shen, Nicolas Gonzalez, **Sinian Zhang**, et al. Advancing the Use of Longitudinal Electronic Health Records: Tutorial for Uncovering Real-World Evidence in Chronic Disease Outcomes. *Journal of Medical Internet Research* 27 (2025): e71873. [Paper](#)
8. Yongkang Xiao, **Sinian Zhang**, Huixue Zhou, Mingchen Li, Han Yang, Rui Zhang. FuseLinker: Leveraging LLM's Pre-trained Text Embeddings and Domain Knowledge to Enhance GNN-based Link Prediction on Biomedical Knowledge Graphs. *Journal of Biomedical Informatics* 158 (2024): 104730. [Paper](#)
9. Jue Hou, Rachel Zhao, Jessica Gronsbell, Yucong Lin, Clara-Lea Bonzel, Qingyi Zeng, **Sinian Zhang**, et al. Generate Analysis-Ready Data for Real-world Evidence: Tutorial for Harnessing Electronic Health Records With Advanced Informatic Technologies. *Journal of Medical Internet Research* 25 (2023): e45662. [Paper](#)
10. Shiyao Huang, Junliang Lyu, **Sinian Zhang**, Ruiying Tang, Huan Xiao, Yuanyuan Zhang, Xiaoling Lu. A Post-processing Machine Learning for Activity Recognition Challenge with OpenStreetMap Data. *Adjunct Proceedings of UbiComp/ISWC 2023*, pages 557–562. [Paper](#)

Preprints & Under Review

1. Xin Zhou*, **Sinian Zhang***, Zhanyan Yang, Mingyuan Xu, Molei Liu, Doudou Zhou. Human-Anchored Inference for Ranking New Models with Large Language Model Judges. *Under review*. arXiv preprint: [2609.19599](#), 2026.
2. **Sinian Zhang**, Chongwei Chen, Jiandong Chen, Lingjie Su, Rui Zhang, Ramya Korlakai Vinayak, Ju Sun. Metric-Aligned Confidence for Selective Medical Image Segmentation. *Under review*, 2026.
3. Shruthi Venkatesh*, **Sinian Zhang***, Wen Zhu, Michele Morris, Rocco Mercurio, Sarah B. Berman, Hansruedi Mathys, Abby L. Olsen, C. Elizabeth Shaaban, Shyam Visweswaran, Oscar L. Lopez, Tianxi Cai, Jue Hou, Zongqi Xia. Predicting the Timing of First Sustained Cognitive Worsening in Alzheimer's Disease Using Real-World Clinical Data and Machine Learning. *Under review*. *medRxiv* preprint: [10.64898/2026.06.02.26354764](#), 2026.
4. **Sinian Zhang***, Kaicheng Zhang*, Ziping Xu, Tianxi Cai, Doudou Zhou. Generalized Linear Markov Decision Process. *Under review*. arXiv preprint: [2506.00818](#), 2025.
5. Jiajun Liang, Yue Liu, Doudou Zhou, **Sinian Zhang**, Junwei Lu. The Wreaths of KHAN: Uniform Graph Feature Selection with False Discovery Rate Control. *Under review*. arXiv preprint: [2403.12284](#), 2024. [Code](#)

SOFTWARE

IntegMultiReg – R package (MCMC sampler in C) 2026
Integrative Bayesian multi-regression of multi-platform biomarkers, supporting continuous, binary and survival outcomes with missing platforms.
[CRAN](#)

PyNILE & RNILE – Python and R wrappers 2025
Wrappers around the NILE Java library for clinical narrative extraction; development guided as a mentor.
[Python](#) [R](#)

TALKS & PRESENTATIONS

Talks

1. **7th ICSA-Canada Chapter Symposium**, McGill University, Montreal, Canada Aug 2026
Federated Adaptive Causal Estimation with High-Dimensional Covariates (FACE-HD)

Conference Abstracts

1. Shruthi Venkatesh, **Sinian Zhang**, Oscar Lopez, Tianxi Cai, Jue Hou, Zongqi Xia. Predicting the Timing of Cognitive Decline in Alzheimer's Disease Using Real-world Clinical Data and Machine Learning (P10-12.009). *Neurology* 106 (11_Supplement_1), 2026. [10.1212/wnl.000000000000217686](https://doi.org/10.1212/wnl.000000000000217686)

TEACHING & MENTORING

Research Mentor – Shuheng Kong & Conglin Ruan, master's students May 2025 – Sep 2025
Guided development of Python and R versions of the NILE package for natural language processing.

Teaching Assistant, Data Science Practice *Renmin University of China*
Beijing, China Jan 2024 – Jun 2024

Teaching Assistant, Probability Theory *Renmin University of China*
Beijing, China Aug 2023 – Dec 2023

Research Mentor – Brittany Wang, high school student Jun 2023 – Sep 2023
Supervised research using real-world data to support an acute ischemic stroke (AIS) study.

ACADEMIC SERVICE

Conference Reviewer: ICLR 2027, AAAI 2027, NeurIPS 2026, ML4H 2026, ICLR 2026

Journal Reviewer: Journal of Intelligent Medicine and Healthcare

HONORS & AWARDS

- **1st Place in the 7th Annual UMN Interdisciplinary Health Data Competition, \$3,000** – University of Minnesota, Twin Cities, 2026
- **Dean's PhD Scholars Award, \$5,000** – University of Minnesota, Twin Cities, 2024
- **Scholarship for Academic Excellence** – Renmin University of China, 2023